

PORTO ALEGRE, Brazil

WMO: 839710

Lat: 29.994S

Lon: 51.171W

Elev: 3

StdP: 101.28

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.2	6.0	1.6	4.2	8.4	3.0	4.7	8.7	10.2	13.0	9.1	14.0	1.3	300	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	34.9	24.6	33.1	24.0	31.9	23.7	26.3	31.6	25.6	30.6	25.0	29.4	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.1	28.3	24.2	19.1	27.4	23.9	18.7	27.1	82.0	31.9	78.9	30.7	76.2	29.5	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.2	7.2	1.5	37.5	1.4	1.3	0.5	38.4	-0.3	39.2	-1.1	40.0	-2.1	40.9		
(5)			1.5	27.8	1.3	1.0	0.5	28.5	-0.2	29.1	-1.0	29.6	-1.9	30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	25.5	25.4	24.0	21.1	17.5	15.2	14.7	16.3	17.6	20.1	22.1	24.3
	DBStd	4.93	2.33	2.43	2.54	3.15	3.24	3.81	3.99	3.94	3.32	2.75	2.70	2.68
	HDD10.0	12	0	0	0	0	1	3	7	2	0	0	0	0
	HDD18.3	455	0	0	1	11	54	107	125	89	52	13	3	1
	CDD10.0	3771	480	431	434	334	234	161	152	197	227	313	364	444
	CDD18.3	1172	222	198	177	95	29	14	12	26	29	68	117	186
	CDH23.3	8731	1848	1578	1252	565	152	102	113	246	209	379	809	1477
	CDH26.7	3078	711	587	418	170	31	23	27	88	76	109	270	569
Wind	WSAvg	3.1	3.5	3.3	3.2	2.8	2.5	2.3	2.6	2.8	3.3	3.6	3.7	3.7
Precipitation	PrecAvg	1389	112	112	108	101	107	133	137	131	144	130	102	108
	PrecMax	1838	238	251	303	295	440	366	309	330	298	307	288	224
	PrecMin	712	10	23	15	0	2	7	19	12	15	20	8	30
	PrecStd	235	53	48	57	63	75	75	69	70	67	76	61	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.1	36.2	35.0	33.1	30.0	29.1	29.8	32.2	33.2	33.1	34.2	36.8
		MCWB	25.1	25.9	24.7	24.0	22.3	21.1	19.5	20.1	21.4	23.1	23.0	24.5
	2%	DB	34.0	33.9	32.5	30.9	27.0	26.8	26.8	29.2	28.2	29.3	31.9	33.9
		MCWB	24.6	25.4	24.3	23.3	20.4	20.3	19.0	19.8	20.6	22.1	22.6	23.9
	5%	DB	32.1	31.9	30.8	28.7	24.9	23.2	23.6	26.0	25.1	27.1	29.8	31.9
		MCWB	24.2	24.6	23.7	22.3	19.8	18.8	18.2	19.3	19.6	20.9	21.6	23.3
	10%	DB	30.4	30.1	29.0	26.5	22.9	20.9	20.8	22.9	22.9	25.1	27.8	29.9
		MCWB	23.8	24.1	23.1	21.4	19.0	17.9	17.3	18.0	18.5	20.1	21.1	22.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.9	27.6	26.7	25.4	23.1	22.0	21.1	22.0	22.9	24.5	25.2	26.6
		MCDB	32.6	33.3	31.3	30.5	27.8	27.2	26.5	27.9	29.2	29.6	30.2	32.4
	2%	WB	26.0	26.4	25.6	24.3	21.6	20.7	19.8	20.6	21.3	23.1	23.8	25.5
		MCDB	31.8	31.6	29.9	28.5	24.7	24.1	24.1	26.1	26.2	27.8	28.8	30.7
	5%	WB	25.2	25.7	24.6	23.3	20.6	19.5	18.7	19.6	20.3	22.0	22.9	24.6
		MCDB	30.1	30.2	28.6	26.8	23.3	22.7	22.5	24.8	24.5	25.7	26.9	29.3
	10%	WB	24.5	24.9	23.9	22.4	19.6	18.4	17.8	18.6	19.4	21.1	22.1	23.8
		MCDB	28.8	28.8	27.4	25.2	22.1	20.6	20.7	22.5	22.3	23.9	26.0	28.2
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.2	9.4	9.5	9.0	9.3	9.6	10.3	9.2	9.0	10.1	10.2
		MCDBR	12.2	11.5	12.0	12.6	12.1	13.3	14.2	16.0	14.1	12.8	13.7	13.6
		MCWBR	4.5	4.4	4.7	5.4	6.0	7.1	7.1	7.6	6.8	6.0	5.6	5.2
	5% WB	MCDBR	10.2	10.0	9.8	9.7	9.4	11.3	11.4	13.6	11.6	10.1	10.6	11.1
		MCWBR	4.3	4.4	4.4	4.7	5.3	6.3	6.5	6.9	6.0	5.3	5.1	4.9
Clear-Sky Solar Irradiance	taub		0.408	0.411	0.400	0.381	0.367	0.382	0.388	0.476	0.522	0.453	0.414	0.412
	taud		2.416	2.409	2.431	2.447	2.459	2.416	2.363	2.060	1.949	2.206	2.340	2.373
	Ebn at Noon		934	912	889	854	815	771	781	741	760	863	922	933
	Edh at Noon		125	122	113	100	89	89	97	145	179	148	134	131
All-Sky Solar Radiation	RadAvg		6.40	5.86	4.95	3.91	2.88	2.43	2.67	3.31	3.99	4.96	6.20	6.53
	RadStd		0.45	0.38	0.34	0.31	0.27	0.18	0.31	0.28	0.39	0.54	0.65	0.52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page